

CellorNuc Starter Quiz: Is This Droplet a Cell (sc) or a Nucleus (sn)?

0. Background

Single-cell RNA-seq (scRNA-seq, hereafter **sc**) profiles intact cells, so the transcripts are dominated by mature cytoplasmic mRNA. Single-nucleus RNA-seq (snRNA-seq, hereafter **sn**) profiles only the nucleus, so unspliced pre-mRNA, intronic reads and nuclear-retained long non-coding RNAs make up a larger share.

Scanpyplus.CellorNucEM is a classifier that predicts whether a droplet is a cell or a nucleus.

Before starting, read the docstring and source of Scanpyplus (<https://github.com/Peng-He-Lab/ScanpyPlus.git>).

1. The data

There are two datasets, each provided as a **full** version and a **6k subsample**. Content and column names are identical; only the number of cells differs. **If your machine is short on memory or things run too slowly, use the 6k version.** You can download the data from this link (<https://ucsf.box.com/s/o6ml14t3swhrjbj2sancufc3k2v4levw>).

Purpose	File	Size
A. Mixed data (cells + nuclei)	mixed_mBDRC_5pCells_multiomeNuclei_full.h5ad	7,813 cells + 53,799 nuclei, 571 MB
	mixed_mBDRC_5pCells_multiomeNuclei_6k.h5ad	3,000 cells + 3,000 nuclei, 63 MB
B. Single-modality data (sc only)	single_KidneyRaji_sc_full.h5ad	119,727 cells, 948 MB
	single_KidneyRaji_sc_6k.h5ad	6,000 cells, 49 MB

Q1. Run the classifier on the mixed dataset A (cells + nuclei)

Download the dataset A and run CellorNucEM on the mixed dataset A.

- **1a.** Are any cell droplets called Typical Nucleus (SN), or any nucleus droplets called Typical Cell (SC)? How well does the signature work on this independent dataset?
- **1b.** Look at the distribution of obs['modality_counts'] in cells vs nuclei and, together with the parameter min_counts=10, explain where these Neutral calls come from. What are the probable reasons for the Neutral calls?

Q2. Run on the single-modality dataset B (sc only)

Many real projects have only sc (or only sn) data. There, the classifier is used to find droplets of the “other kind” that slipped in — for example, droplets in an sc library whose cytoplasm lysed, leaving only the nucleus.

- **2a.** Load dataset B and run CellorNucEM. What fraction of droplets are Nucleus-like Cell (SC)?
- **2b.** Plot a histogram of `sc_frac`. Is it unimodal or bimodal?

Q3. What do the anchors (constraints) do in the single-modality case?

CellorNucEM has an “anchor” related mechanism:

- **3a.** On dataset B, run (i) default parameters; `init='anchored'`, (ii) `init='split'`. Compare the two component means `p` between the two fits. Are the differences large? If yes, what could be the reasons?
- **3b.** Switch to dataset A: take only droplets with `suspension_type == 'cell'` and compare (i) and (ii) in the same way; then do the same with nucleus only.